

TravelMate AI Project Documentation

Project Title

TravelMate AI – Multi-Agent AI Travel Planner

Project Description

TravelMate AI is a multi-agent AI-powered travel planning system that generates personalized travel itineraries using React, FastAPI, LangGraph, PostgreSQL, Gemini/Claude and external travel APIs.

Problem Statement

Users visit multiple websites to plan trips, making planning time-consuming.

Proposed Solution

Provide flights, hotels, attractions, weather, budget estimation and personalized itineraries on one platform.

System Prompt

You are TravelMate AI. Understand user preferences, recommend flights and hotels, suggest attractions, consider weather, generate a day-wise itinerary, estimate budget and return JSON.

Architecture

User -> React -> FastAPI -> LangGraph -> Memory -> Research -> Flight -> Hotel -> Itinerary -> Budget -> Gemini/Claude -> PostgreSQL -> JSON -> React UI

Pipeline

User Input -> React -> FastAPI -> LangGraph -> AI Agents -> External APIs -> Gemini/Claude -> JSON -> Frontend

Technology Stack

React, Tailwind CSS, FastAPI, Python, JavaScript/TypeScript, LangGraph, PostgreSQL, Render, Gemini/Claude

Demo URL

<https://your-demo-link.com>

GitHub

<https://github.com/your-username/TravelMate-AI>

Future Scope

Voice assistant, live booking, multilingual support, offline mode, collaborative planning.